Business Frameworks
Term 1_Mid Term Examination

Total Time: 60 Minutes I Total Marks: 30

Modernizing Gupta Kirana Store

Background & Current State

Gupta Kirana Store is an independent, family-run neighborhood grocery shop operating in a high-density residential area of Bangalore. The store has been in business for over 15 years and has a loyal walk-in customer base within a 500-meter radius.

Despite steady daily footfall, revenue growth has stagnated over the past three years.

Key Operational Constraints:

- **No Home Delivery:** All sales are strictly over-the-counter / walk-in.
- **Manual Record Keeping:** Sales, payments, and supplier ledgers are tracked manually on paper. There is no Digital Point of Sale (POS), digital inventory tracking, or SKU-level sales data.
- **No Remote Ordering:** The store does not accept orders over WhatsApp, phone, or messaging apps.
- **Limited Floor Space:** Shelves are packed, making it hard to introduce high-margin impulse items without removing existing stock.

The owner, Mr. Ramesh Gupta, believes that since customers know where the store is, digital tools and delivery options are an unnecessary expense and operational hassle.

Questions for you

1. **Hypothesis Generation: Formulate 3 hypotheses** (2 for growth & 1 for profitability) on how introducing the following, can expand Gupta Kirana Store's business, and help it become more profitable
    a. WhatsApp ordering and home delivery,
    b. Digital sales/inventory tracking
2. **Hypothesis Testing & Problem Solving:** Describe how Mr. Gupta can run low-cost pilots to test each hypothesis. Based on the experimental sample data provided below, analyze whether each hypothesis is supported & validated.

3. **The Growth Flywheel:** Construct a strategic growth flywheel showing how these advancements potentially reinforce each other to create continuous, self-fulfilling business growth.

Experimental Data

Mr. Gupta ran a 4-week pilot test with 50 opt-in regular households in his immediate neighborhood. Below is the data collected:

Metric	Pre-Pilot Baseline (Manual / Walk-in Only)	4-Week Pilot (WhatsApp + Delivery + POS Tracking)
Average Order Value (AOV)	₹280	₹540 (via WhatsApp orders) / ₹310 (walk-ins)
Monthly Purchase Frequency / Household	3.2 walk-in orders/month	3.4 WhatsApp orders + 1.7 walk-in orders per month
Stock-Out Rate on Key Essentials	18% (estimated out-of-stock instances)	4% (tracked via digital inventory logging)
Customer Retention (Repeat Monthly Rate)	68%	86% for pilot participants, 68% for non-pilot participants
Delivery Cost per Order	N/A	₹25 (using a part-time delivery runner)
Contribution Margin per Order (Revenue - COGS)	12%	15% (higher sales of high-margin packaged snacks & personal care items)